

Designing PhenoProbe: An Interview Tool for Inquiring into the Phenological Knowledge of Taiwanese Elders

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ABSTRACT

With the growing urgency of addressing climate change, HCI and interaction design fields are seeking novel yet practical solutions. Since weather patterns undergo long-term changes, elders' phenological knowledge and everyday experiences can offer valuable insights. Nonetheless, these insights need to be adequately represented in sustainability discussions. We present the research-through-design process behind PhenoProbe, an interview tool that employs a series of cards inspired by the 24 solar terms. PhenoProbe incorporates Taiwanese seasonal soundscapes with surreal visuals generated through proverb prompts to cultivate meaningful conversations between elders and researchers. The tool is crafted to encourage elders to comfortably engage in and reflect upon their daily weather-related experiences. Our work delivers two main contributions: 1) A tool that engages elders in discussions about phenological patterns and embodied experiences, and 2) An approach that weaves familiar cultural elements into open-ended elicitation materials, fostering the involvement of older adults in the design process.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Systems and tools for interaction design;

KEYWORDS

Elders, Phenological Knowledge, Elicitation Interview, 24 Solar Terms, Cultural Probe

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1 INTRODUCTION

In the field of Human-Computer Interaction (HCI), there is a growing emphasis on sustainability [2, 8] that encompasses various

domains such as environmental [5], botanical [21], animal [1], and household essentials [11]. Understanding weather patterns is significantly challenging due to their complexity and vast scale, often surpassing human perception spatially and temporally. Timothy Morton's concept of "hyperobjects" [17] accurately characterizes this phenomenon, highlighting weather patterns extending beyond our perceptual capabilities.

While humans find it challenging to comprehend climate as a complex, overarching entity, studying the 24 *solar terms* (Taiwanese Mandarin pronunciation: 'Jie-Chi') reveals the patterns of extreme climate changes across history. We consider the connection between the lived experiences of the elderly and the concept of Jie-Chi. In times when technology was less advanced, elders relied on their recollections, experiences, and sensory perceptions to gather insights on navigating climate shifts, directly engaging with their surroundings. Thus, we posit that incorporating the insights from Jie-Chi and the perspectives of the elderly can unveil novel opportunities for research into climate-related phenomena. However, research focusing on how they adapt to weather changes is limited, often overlooking their creative and practical skills and emphasizing their dependency needs. To address this gap, our research explores two main questions: 1) How do elders interact with the weather in their daily lives, and 2) how can we deepen our understanding of the human-weather relationship through eliciting their experiences? To investigate, we used probing-inspired [9, 18, 23] and card-based method [13] to delve into elders' experiences with the weather. We identified the Jie-Chi as the key concepts to facilitate our inquiry. Jie-Chi is derived from a traditional Chinese calendar system and reflects how agricultural communities historically interpreted and adapted to weather changes. Jie-Chi symbolizes the connection between humans, the environment, and socio-cultural practices, linking routine activities to the natural world. Although the relevance of such traditional knowledge in the modern context might be questioned, previous studies [4] have successfully applied traditional farmer almanacs in developing innovative smart home solutions. This suggests that blending traditional and contemporary elements can ignite creativity in design. Building on these insights, we developed *PhenoProbe*, an interview tool combining proverbs of Jie-Chi, fictional weather illustrations, and local soundscapes. The toolkit includes 24 Jie-Chi cards and a sound-playing device, enabling us to conduct in-depth interviews with 19 elders. By focusing on the design process of the PhenoProbe, the paper contributes in two ways: 1) by demonstrating a tool that can encourage older people to reflect on their daily relationship with the weather, and 2) by providing insights for weaving familiar cultural elements into

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Research Process

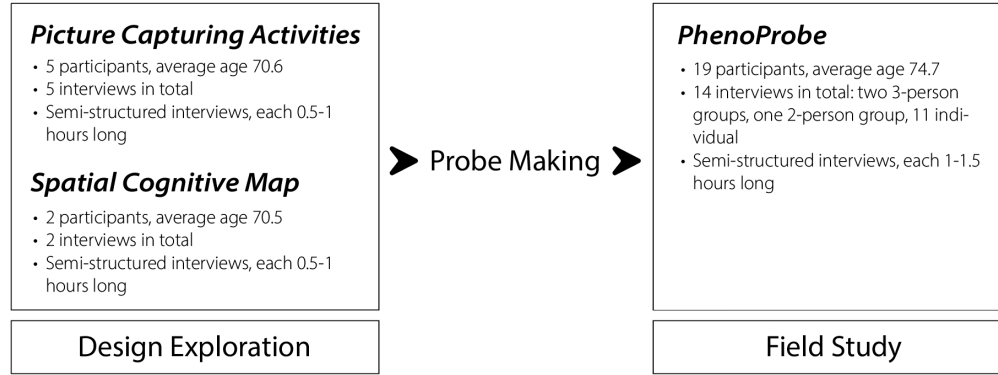


Figure 1: The research-through-design process of PhenoProbe

open-ended elicitation materials to foster the involvement of elders in the design process.

2 RESEARCH PROCESS

Our study adopted a research-through-design (RtD) approach [14]. Building on the principle of RtD, we anticipated that research findings should naturally evolve rather than be preconceived. Adopting this approach allowed our design goals to be flexible and developed as we navigated the challenges inherent in the research process. The process (Fig.1) was organized into three main stages: *Design Exploration*, *Probe Making*, and *Field Study*. The initial stage focused on sampling Taiwanese elders' life experiences and habits through two cultural probes, providing insights for the later design. In the second stage, we employed conceptual, visual, and auditory materials for designing PhenoProbe. The design incorporated traditional proverbs, fictional weather illustrations, and local soundscapes into a tangible toolkit. In the final stage, these elicitation materials of PhenoProbe were brought into the interviews with elders to facilitate engaging conversations, helping the elders confidently articulate their weather-related experiences. We applied thematic analysis[3] to extract insights from the transcribed interviews and built up our findings. Maintaining a structured yet flexible approach throughout these stages, we could critically assess the lessons learned from our research process, remaining receptive to unforeseen discoveries. This approach gave us a more profound understanding of the elders' perceptions and imaginations concerning interaction with the weather.

3 DESIGN EXPLORATION

Our research is focused on understanding how Taiwanese elders incorporate weather into their daily lives. The first design exploration stage included two essential activities: *Picture-Capturing* and *Spatial-Cognitive Mapping*. These activities aimed to explore how the elders interact with their surroundings with their body and mind. This would help us gain valuable insights and inspiration for our further design process.

3.1 Picture Capturing Activities

Our *Elderly-Meme Probe* incorporated an activity that engages elderly people in exploring their daily routines by taking pictures. Inspired by Gaver's cultural probe [9], our initiative aimed to generate inspirational data that stimulates designers' inspiration. Therefore, we focused on understanding the elders' personal preferences and beliefs rather than just collecting their objective opinions. To achieve this, we designed the Elderly-Meme Probe with elements of ambiguity [10], blending playful and thought-provoking questions. When materials possess ambiguous qualities, it can result in messages that are difficult to comprehend, mysterious, and subject to varying interpretations. Such situations often prompt people to seek clarity on the unclear aspects of the message actively. To create ambiguous materials, we incorporated the popular "Elder Meme" format and supplemented it with unconventional and absurd queries to direct elders in their daily photo-taking tasks. This approach provided elders with intellectually stimulating material that is entertaining and captivating, fostering a renewed outlook on everyday life issues. Participants were invited to select a card daily and take a photograph corresponding to the theme or suggestion on the card. All five elderly participants had prior experience



Figure 2: (left) Elderly-Meme Task Cards: These cards combine everyday tasks with positive messages, such as "May good fortune be with you! Have a wonderful weekend! Take a picture of something that represents good luck, wishing you prosperity in all areas of life". (right) The Elderly-Meme Probe includes an instant camera, 11 task cards, two pens, and various stickers to assist elders in documenting their activities.

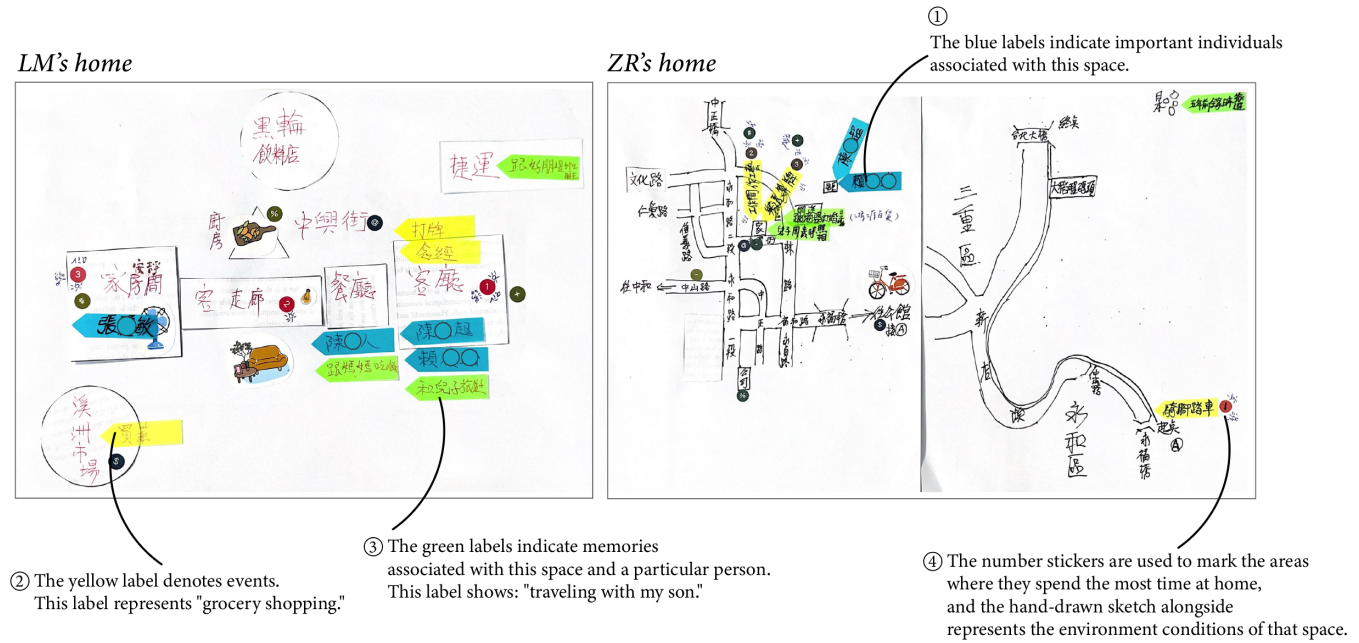


Figure 3: (left) LM's illustration centers on her house, with three surrounding locations: a subway station, a traditional market, and a snack stand. (right) ZR's illustration, in contrast, highlights his range of motion by beginning from his house and extending into a comprehensive network that covers an entire region.

with digital devices, including computers, smartphones, and digital cameras.

The Materials: Elderly-Meme Probe collected paper-based task cards (Fig.2-left) that draw inspiration from the "Elder Memes" originating from the elderly community in Taiwan. These memes feature everyday imagery accompanied by greetings, blessings, or motivational phrases. We collected sixty memes from elders' communities and online groups to better understand their appeal and usage. Our analysis revealed that these memes serve as a means of fostering inter-generational care expression, conversation initiation, and social connection. With themes ranging from greetings to faith, holidays, blessings, and everyday dialogues, we created daily photography tasks based on these themes, incorporating elder memes' visual and narrative style with playful narrative alterations to maintain the user's engagement. To make these tasks more accessible for elders, we provided eleven meme-inspired task cards, an instant camera, pens, and stickers (Fig.2-right).

The Task: To ensure that the elderly participants with varying technological skills could easily use the photography equipment provided, we offered them a choice between two types of cameras: a tangible instant camera or their mobile phone. The participants were split into two groups, with two groups selecting the instant camera option and the remaining three opting for their smartphones. We then tailored the method for documenting thoughts to their chosen device. Participants using instant cameras would write their thoughts directly onto the pictures with pens and stickers. On the other hand, those who preferred smartphones had the convenience of recording their thoughts directly on their phones.

Reflection: Our activity discovered that many elderly individuals incorporate the 'Jie-Chi' into their daily routines in various ways. Jie-Chi was initially designed to guide agricultural practices. Each of the 24 Jie-Chi corresponds to specific weather patterns and cultural traditions. We found a significant correlation between Jie-Chi and the daily activities of the elders, such as the ritual of preparing pomelos for friends during the Mid-Autumn Festival, which is on the date of White Dew (starting from September 7-9 each year). The elders also strongly connect with nature and align their activities of daily living with weather patterns, such as airing quilts before the Winter Solstice (starting from December 22 each year). Additionally, the elders discuss weather and natural shifts, rooted in Jie-Chi, to express their care for younger generations. Essentially, Jie-Chi extends beyond the general concept of weather, encompassing broader aspects of life, including personal habits, natural interaction, family bonds, and social customs.

3.2 Spatial Cognitive Map

This probing study examined the intricate connection between elders and their living spaces through a spatial exploration activity. To guide our investigation, we drew inspiration from Stea's book [7] on cognitive mapping, which underscores the importance of organizing information about one's surroundings. The objective was to understand better how the elders perceive and interact with their living spaces. During a one-hour cognitive mapping session, we asked the elderly participants to visually map and discuss their living environments. The study recruited two participants: ZR, a retired government employee with a mathematics background who



Figure 4: (left) One side of the card features Jie-Chi proverbs. (right) A series of Jie-Chi cards, consisting of 24 individual cards, each presenting a Jie-Chi name alongside its fictional illustration.

enjoys cycling, and LM, who is skilled in cooking, managing home affairs, and socializing while playing Mahjong.

The Materials: Recent research has revealed that elderly individuals engage more naturally and intuitively in the design process when utilizing simple, low-tech materials such as paper, sticky notes, and colored pencils [15, 19]. We provided various easy-to-use creative tools to facilitate their participation in creating spatial cognitive maps. These tools included a selection of pens, assorted labels, sticky notes, geometrically shaped papers, and various stickers representing different objects.

The Task: We structured three distinct themed drawing tasks for our elderly participants, with each task to be executed on A4 white paper, as shown in Fig.3. The first theme was to illustrate their living environment. The second task involved identifying and labeling the people (Fig. 3-1), memories (Fig. 3-3), and activities (Fig. 3-2) in different areas of their space. Finally, for the third task, we asked the participants to depict and annotate the physical feelings and sensory experiences (Fig. 3-4) they encountered in their living spaces, such as the feel of consistent sunlight. This approach aimed to collect detailed environmental descriptions, focusing primarily on weather-related aspects.

Reflection: Through our study of elderly participants from diverse backgrounds, we discovered that their perceptions of the same living space varied greatly. However, we found that their feedback predominantly centered around objective details related to physical locations and phenomena, with limited insight into their daily routines and interactions with nature. We realized that this activity unintentionally steered them towards focusing more on spatial accuracy rather than expressing their lived experiences in those environments. We aim to extract design inspiration from their personal experiences, so we analyzed the collected information and explored alternative conceptual approaches to encourage more profound discussions about their spatial relationships. Research [6, 20] indicates that sound can elicit emotional and cognitive responses among the elderly; besides, its abstract nature compared to visual elements makes it a valuable tool for nuanced interpretations of space. As a result, we intend to conduct further research into using sound to stimulate reflection and imagination among the elderly concerning their living environments.

4 PHENOPROBE

We created an interview tool, PhenoProbe, which aims to facilitate elderly individuals in sharing their weather-related experiences and

insights. We understood the significance of Jie-Chi; therefore, we incorporated it into the PhenoProbe with ambiguity. This integration is manifested in a set of 24 Jie-Chi cards (Fig.4), each dedicated to a different Jie-Chi. One side of each card presents a Jie-Chi proverb, while the other features an artistic illustration. Additionally, each card is associated with three unique ambient sounds that reflect its respective Jie-Chi. The PhenoProbe comprises these 24 Jie-Chi cards, a collection of 72 soundscapes, and two sets of question cards. The following paragraphs will detail the development process of these components.

Integrating Jie-Chi proverb into the card Our cards were designed to provide participants with information about each Jie-Chi and its corresponding weather patterns. The front of each card features the name of the Jie-Chi. At the same time, the back showcases a carefully curated Jie-Chi proverb (Fig. 4-left) that sheds light on environmental transformations and essential details specific to that solar term. These proverbs were meticulously selected from books and online resources to reflect Taiwan’s local climate nuances accurately. For example, the proverb *Rain on the Lantern Festival, sun before Qingming* highlights the correlation between Taiwan’s weather and cultural practices during the Qingming period (April 4th or 5th each year). Our card tool includes 24 proverbs, each resonating with a unique aspect of Taiwan’s climate.

Creating imaginary weather illustrations inspired by local proverbs The design of the card’s illustration (Fig.4-right) was intended to kindle the participants’ imagination about the Jie-Chi and encourage them to explore their connection with weather patterns. This technique facilitated the rapid production of various innovative images while also allowing for the customization of their degree of fictionalization to better align with the researchers’ expectations. Additionally, this approach enhanced the efficiency of stimulating discussions and engagement among the participants. We utilized the generative AI tool, Stable Diffusion, for crafting images. We established a prompt template for image generation: *"A [type of picture] of a [main subject], [style cues]"*. For the type of picture, we uniformly selected ‘digital illustrator’ and enriched the images with imaginative and surrealistic style tags like surreal and pop art to enhance their fictional aspect. In defining the main subject, we extracted key nouns from Jie-Chi proverbs to form thematic prompts. For example, we summarized the proverb *"Rain strikes the lanterns during the Lantern Festival, the sun shines before Qingming"* as the theme: *"Lantern Festival at the Rain Water season"*. To maintain a consistent style across all images, our style

Interview Process

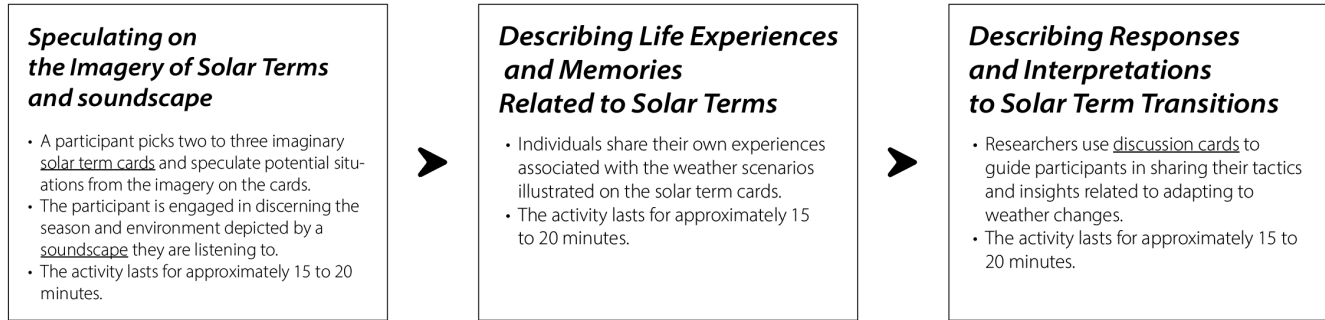


Figure 5: The interview process with PhenoProbe

cues included uniform visual styles, resolution, and color schemes: 4K resolution, detailed richness, styles popular on Artstation, and a fantasy color palette. A total of 276 visuals were generated, of which 24 were carefully chosen to feature on our cards.

Employing soundscapes within the tool Recent research [6, 20] suggests that auditory experiences can deeply cognitively and emotionally engage individuals. Our prior explorations have led us to consider that abstract auditory experiences may be more effective than visual experiences for inspiring the elderly to re-explore their surroundings. To this end, we have compiled sounds from various Taiwanese landscapes corresponding with Jie-Chi. Our collection comprises a mix of audio from the Taiwan Sound Database, YouTube, and field recordings, intending to provide the elderly with familiar everyday sounds that resonate with them cognitively and emotionally. In total, we collected 94 sounds and selected 72 for the sound database of our Jie-Chi cards. The sounds are categorized into natural and cultural groups and are evenly distributed across 24 Jie-Chi themes. Each card is designed to trigger three different sounds, including at least one cultural and one natural soundscape. To facilitate easy interaction for the elderly, we've designed a device that plays these soundscapes using RFID technology combined with a DFPlayer. Each card carries a unique RFID tag. When an elderly individual brings a card close to the device, it recognizes the tag and randomly plays one of the three corresponding ones through the DFPlayer, connected to an 8-ohm speaker.



Figure 6: (left) The kit of PhenoProbe. (middle and right) The scenario of participants using PhenoProbe.

4.1 Interview with PhenoProbe

During an interview, the participants were engaged in three key activities (Fig.5). Firstly, they speculated the weather scenarios that the solar term cards and soundscape represent. Secondly, they shared their personal experiences related to these solar terms. Finally, in the last stage, participants and researchers discussed strategies and experiences in handling different weather scenarios, utilizing topics from discussion cards.

4.2 Participants

Throughout the study, researchers conducted interviews with a total of 19 elderly participants, with 10 from an elder community in Yunlin, 6 from Tainan, and 3 from Taipei City. The average age of the participants was 74.7. The study format included three group discussions: two groups consisted of three participants each, and the third group was a married couple. In addition to the group discussions, individual interviews were also conducted. Each interview session, which always involved at least one researcher, lasted an average of 1 to 1.5 hours.

5 PRELIMINARY RESULTS AND FINDINGS

The PhenoProbe is effective in encouraging elderly participants to share their weather-related experiences, revealing valuable insights. This tool involved older individuals in conversations about the effect of weather variations on their everyday lives. Preliminary outcomes of their interactions with the Jie-Chi cards and soundscapes are as follows:

Jie-Chi Card: 1) The Jie-Chi illustration prompted recollections of similar scenes that the elderly had experienced in the past. 2) The Jie-Chi illustration caused people's physical sensations, such as a set of cotton images making them feel cool. 3) The Jie-Chi illustration confused the elderly, as some participants commented on scenes with three suns, stating that they had visited a similar place in the past. 4) Due to unfamiliarity with some proverbs, the elderly combined the proverbs, solar term names, and images to reinterpret them into a set of explanations that made sense to them.

Soundscape: 1) The soundscape evoked memories from the elderly participants. 2) When encountering new sounds, the elderly

reconstructed the definition of the soundscape from their memories. Overall, the Jie-Chi illustration, solar term proverbs, and local soundscapes associated with the solar terms sparked the interest of the elderly, prompting them to actively participate in the activity and explore their various interactions with the weather. These experiences were not limited to descriptions of weather conditions but also included memories, imagined scenarios, and even real physical sensations.

6 DISCUSSION & FUTURE WORK

This study presents an interview tool that combines Jie-Chi proverbs, fictional weather illustrations, and local soundscapes to effectively engage the elderly participants and stimulate their curiosity during discussions with researchers. The tool generates open-ended activities that inspire exploration and self-discovery, promoting active participation and empowering the elderly to express themselves freely without fear of being judged. Moreover, the open-ended nature of these topics not only assists researchers in adopting a "co-participatory" role [22], but also fosters a collaborative environment. As there are no definitive answers to open-ended questions, these scenarios enable both researchers and the elderly to get engaged in joint discussions, transforming the researchers from passive listeners to active participants in conversations. This collaborative approach treats elderly individuals as equals and avoids a paternalistic stance from researchers, challenging stereotypes of fragility and dependence [12].

Nonetheless, Lindsay et al. [16] expressed reservations about the potential difficulties the elderly may encounter in conceptualizing their ideas and outcomes due to the unstructured nature of these activities, which could result in a sense of loss of control during the process. To address this, familiarizing the design materials could be a viable strategy to help the elderly connect the topics to their own life experiences. For example, in PhenoProbe, a combination of actual (Jie-Chi proverbs and soundscapes) and imaginary components (fictional illustrations of Jie-Chi), can be used to create recognizable features in the open-ended activities that motivate exploration and self-discovery. This approach encourages active engagement and enables elderly individuals to discover personal significance in unfamiliar situations. To further understand the experience of our users and evaluate the effectiveness of our tool, we are currently conducting in-depth experiments and interviews. As part of our future studies, we aim to incorporate the invaluable wisdom of the elderly demographic into the development of intelligent weather objects. This inclusive approach will enable us to comprehensively explore the practical applications of interaction design for weather adaptation, considering diverse perspectives within a more applicable framework.

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REFERENCES

- [1] Heidi R. Biggs, Jeffrey Bardzell, and Shaowen Bardzell. 2021. Watching Myself Watching Birds: Abjection, Ecological Thinking, and Posthuman Design. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 619, 16 pages.
- [2] Eli Blevis. 2007. Sustainable interaction design: invention & disposal, renewal & reuse. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 503–512.
- [3] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [4] Liang-Yu Chen, Ji-Hong Huang, Yu-Hao Lee, Chia-Hsu Huang, and Rung-Huei Liang. 2019. A world following farmer almanac: speculation on lifestyle interweaving folk religion and smart home. In *Companion Publication of the 2019 on Designing Interactive Systems Conference 2019 Companion*. 147–151.
- [5] Aykut Coskun, Nazli Cila, Iohanna Nicenboim, Christopher Frauenberger, Ron Wakkary, Marc Hassenzahl, Clara Mancini, Elisa Giaccardi, and Laura Forlano. 2022. More-than-human Concepts, Methodologies, and Practices in HCI. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–5.
- [6] Smit Desai, Morgan Lundy, and Jessie Chin. 2023. "A Painless Way to Learn." Designing an Interactive Storytelling Voice User Interface to Engage Older Adults in Informal Health Information Learning. In *Proceedings of the 5th International Conference on Conversational User Interfaces*. 1–16.
- [7] Roger M Downs and David Stea. 1973. *Image and environment: Cognitive mapping and spatial behavior*. Transaction Publishers.
- [8] Emily Elhacham, Liad Ben-Uri, Jonathan Grozovski, Yinon M Bar-On, and Ron Milo. 2020. Global human-made mass exceeds all living biomass. *Nature* 588, 7838 (2020), 442–444.
- [9] Bill Gaver, Tony Dunne, and Elena Pacenti. 1999. Design: Cultural Probes. *Interactions* 6, 1 (jan 1999), 21–29.
- [10] William W. Gaver, Jacob Beaver, and Steve Benford. 2003. Ambiguity as a Resource for Design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Ft. Lauderdale, Florida, USA) (CHI '03). Association for Computing Machinery, New York, NY, USA, 233–240.
- [11] Elisa Giaccardi and Elisa Giaccardi. 2020. Casting things as partners in design: towards a more-than-human design practice. *Relating to things: design, technology and the artificial*. Bloomsbury, London (2020), 99–132.
- [12] Masako Kitazaki, Iohanna Nicenboim, and Elisa Giaccardi. 2019. Connected Resources-Empowering Older People to Age Resourcefully. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–1.
- [13] Bowen Kong, Rung-Huei Liang, MengChi Liu, Shu-Hsiang Chang, Hsiu-Chen Tseng, and Chian-Huei Ju. 2021. Neuromancer workshop: Towards designing experiential entanglement with science fiction. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [14] Ilpo Koskinen, John Zimmerman, Thomas Binder, Johan Redstrom, and Stephan Wensveen. 2011. *Design Research Through Practice*. Morgan Kaufmann.
- [15] Tuck Wah Leong and Toni Robertson. 2016. Voicing Values: Laying Foundations for Ageing People to Participate in Design. In *Proceedings of the 14th Participatory Design Conference: Full Papers - Volume 1* (Aarhus, Denmark) (PDC '16). Association for Computing Machinery, New York, NY, USA, 31–40.
- [16] Stephen Lindsay, Daniel Jackson, Guy Schofield, and Patrick Olivier. 2012. Engaging older people using participatory design. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 1199–1208.
- [17] Timothy Morton. 2013. *Hyperobjects: Philosophy and Ecology after the End of the World*. U of Minnesota Press.
- [18] Doenja Oogjes, William Odom, and Pete Fung. 2018. Designing for an other home: Expanding and speculating on different forms of domestic life. In *Proceedings of the 2018 Designing Interactive Systems Conference*. 313–326.
- [19] Alisha Pradhan, Ben Jelen, Katie A. Siek, Joel Chan, and Amanda Lazar. 2020. Understanding Older Adults' Participation in Design Workshops. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–15.
- [20] Daniel C Richardson, Nicole K Griffin, Lara Zaki, Auburn Stephenson, Jiachen Yan, Thomas Curry, Richard Noble, John Hogan, Jeremy I Skipper, and Joseph T Devlin. 2018. Measuring narrative engagement: The heart tells the story. *BioRxiv* (2018), 351148.
- [21] Margrete Lodahl Rolighed, Ester Marie Aagaard, Marcus Due Jensen, Raune Frankjaer, and Lone Koefoed Hansen. 2022. Plant Radio: Tuning in to plants by combining posthumanism and design. In *Designing Interactive Systems Conference*. 666–676.
- [22] Wenn-Chieh Tsai, Daniel Orth, and Elise Van Den Hoven. 2017. Designing memory probes to inform dialogue. In *Proceedings of the 2017 Conference on Designing Interactive Systems*. 889–901.
- [23] Jayne Wallace, John McCarthy, Peter C Wright, and Patrick Olivier. 2013. Making design probes work. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 3441–3450.